

3. SQL Queries for Data Manipulation, Access Control, and Transactions

Design and run SQL queries to demonstrate the following:

a) Data Manipulation (DML): Use SQL statements to INSERT, UPDATE, and DELETE records. Apply arithmetic, logical, set operators, pattern matching, and string functions.

```
mysql> CREATE TABLE Book (Book_ID INT PRIMARY KEY,Book_Name  
VARCHAR(50),Author VARCHAR(50),Price INT,Category VARCHAR(30));
```

Query OK, 0 rows affected (0.08 sec)

```
mysql> CREATE TABLE Issued_Book (Book_Name VARCHAR(50));
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> INSERT INTO Book VALUES(1, 'Database Systems', 'Navathe', 500, 'Education'),(2,  
'Operating System', 'Silberschatz', 650, 'Education'),(3, 'C Programming', 'Dennis Ritchie', 300,  
'Programming'),(4, 'Python Basics', 'Guido', 400, 'Programming');
```

Query OK, 4 rows affected (0.01 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql> show tables;
```

```
+-----+  
| Tables_in_dept |  
+-----+  
| book           |  
| course         |  
| department     |  
| issued_book    |  
| student        |  
+-----+
```

5 rows in set (0.00 sec)

```
mysql> select * from Book;
```

Book_ID	Book_Name	Author	Price	Category
1	Database Systems	Navathe	500	Education
2	Operating System	Silberschatz	650	Education
3	C Programming	Dennis Ritchie	300	Programming
4	Python Basics	Guido	400	Programming

4 rows in set (0.00 sec)

```
mysql> UPDATE Book SET Price = Price + (Price * 0.10)WHERE Category = 'Programming';
```

Query OK, 2 rows affected (0.01 sec)

Rows matched: 2 Changed: 2 Warnings: 0

```
mysql> select * from Book;
```

Book_ID	Book_Name	Author	Price	Category
1	Database Systems	Navathe	500	Education
2	Operating System	Silberschatz	650	Education
3	C Programming	Dennis Ritchie	330	Programming
4	Python Basics	Guido	440	Programming

4 rows in set (0.00 sec)

```
mysql> DELETE FROM Book WHERE Price < 350;
```

Query OK, 1 row affected (0.01 sec)

```
mysql> select * from Book;
```

Book_ID	Book_Name	Author	Price	Category
1	Database Systems	Navathe	500	Education
2	Operating System	Silberschatz	650	Education
4	Python Basics	Guido	440	Programming

3 rows in set (0.00 sec)

APPLY ARITHMETIC OPERATORS

```
mysql> SELECT Book_Name, Price, Price + 50 AS New_Price FROM Book;
```

Book_Name	Price	New_Price
Database Systems	500	550
Operating System	650	700
Python Basics	440	490

3 rows in set (0.00 sec)

APPLY LOGICAL OPERATORS (AND, OR, NOT)

```
mysql> SELECT * FROM Book WHERE Category = 'Education' AND Price > 500;
```

```

+-----+-----+-----+-----+-----+
| Book_ID | Book_Name   | Author   | Price | Category |
+-----+-----+-----+-----+-----+
|      2 | Operating System | Silberschatz | 650 | Education |
+-----+-----+-----+-----+-----+

1 row in set (0.00 sec)

```

PATTERN MATCHING (LIKE Operator)

```
mysql> SELECT * FROM Book WHERE Book_Name LIKE 'D%';
```

```

+-----+-----+-----+-----+-----+
| Book_ID | Book_Name   | Author   | Price | Category |
+-----+-----+-----+-----+-----+
|      1 | Database Systems | Navathe | 500 | Education |
+-----+-----+-----+-----+-----+

1 row in set (0.00 sec)

```

STRING FUNCTIONS

```
mysql> SELECT UPPER(Book_Name) AS Book_Upper, LOWER(Author) AS Author_Lower
FROM Book;
```

```

+-----+-----+
| Book_Upper   | Author_Lower |
+-----+-----+
| DATABASE SYSTEMS | navathe      |
| OPERATING SYSTEM | silberschatz |
| PYTHON BASICS   | guido        |
+-----+-----+

3 rows in set (0.00 sec)

```

```
mysql> SELECT Book_Name, LENGTH(Book_Name) AS Name_Length FROM Book;
```

```

+-----+-----+
| Book_Name | Name_Length |
+-----+-----+
| Database Systems | 16 |
| Operating System | 16 |
| Python Basics | 13 |
+-----+-----+

```

3 rows in set (0.01 sec)

```
mysql> SELECT SUBSTRING(Book_Name, 1, 4) AS Short_Name FROM Book;
```

```

+-----+
| Short_Name |
+-----+
| Data      |
| Oper      |
| Pyth      |
+-----+

```

3 rows in set (0.00 sec)

SET OPERATORS

```
mysql> INSERT INTO Issued_Book VALUES ('Database Systems'), ('Python Basics');
```

Query OK, 2 rows affected (0.01 sec)

Records: 2 Duplicates: 0 Warnings: 0

```
mysql> select * from Issued_Book;
```

```
+-----+
| Book_Name |
```

```
+-----+
| Database Systems |
```

```
| Python Basics |
+-----+
```

2 rows in set (0.00 sec)

```
mysql> SELECT Book_Name FROM Book UNION SELECT Book_Name FROM
Issued_Book;
```

```
+-----+
| Book_Name |
```

```
+-----+
| Database Systems |
```

```
| Operating System |
| Python Basics |
```

```
+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT Book_Name FROM Book INTERSECT SELECT Book_Name FROM
Issued_Book;
```

```
+-----+
| Book_Name |
```

```
+-----+
| Database Systems |
```

```
| Python Basics |
+-----+
```

2 rows in set (0.00 sec)

```
mysql> SELECT Book_Name FROM Book EXCEPT SELECT Book_Name FROM  
Issued_Book;
```

```
+-----+
```

```
| Book_Name |
```

```
+-----+
```

```
| Operating System |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

b) Access Control (DCL): Use GRANT, REVOKE, and ROLE commands to manage user access.

Step 1: Create Database & Table (Admin)

```
mysql> CREATE DATABASE college_db;
```

Query OK, 1 row affected (0.01 sec)

```
mysql> USE college_db;
```

Database changed

```
mysql> CREATE TABLE student (roll_no INT PRIMARY KEY,name VARCHAR(50),marks INT);
```

Query OK, 0 rows affected (0.02 sec)

Insert Data

```
mysql> INSERT INTO student VALUES(1,'Amit',85),(2,'Neha',90),(3,'Ravi',78);
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> select * from student;
```

```
+-----+-----+-----+
```

```
| roll_no | name | marks |
```

```
+-----+-----+-----+
```

```
| 1 | Amit | 85 |
```

```
| 2 | Neha | 90 |
```

```
| 3 | Ravi | 78 |
```

```
+-----+-----+-----+
```

3 rows in set (0.00 sec)

Step 2: Create Users (MySQL)

```
mysql> CREATE USER 'admin1'@'localhost' IDENTIFIED BY 'admin123';
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> CREATE USER 'faculty1'@'localhost' IDENTIFIED BY 'faculty123';
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> CREATE USER 'student1'@'localhost' IDENTIFIED BY 'student123';
```

Query OK, 0 rows affected (0.01 sec)

Step 3: Create ROLES (MySQL 8.0+)

```
mysql> CREATE ROLE 'admin_role';
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> CREATE ROLE 'faculty_role';
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> CREATE ROLE 'student_role';
```

Query OK, 0 rows affected (0.01 sec)

Step 4: GRANT Privileges to ROLES

Admin Role – Full Access

```
mysql> GRANT ALL PRIVILEGES ON college_db.* TO 'admin_role';
```

Query OK, 0 rows affected (0.01 sec)

Faculty Role – Read & Update Marks

```
mysql> GRANT SELECT, UPDATE ON college_db.student TO 'faculty_role';
```

Query OK, 0 rows affected (0.01 sec)

Student Role – Read Only

```
mysql> GRANT SELECT ON college_db.student TO 'student_role';
```

Query OK, 0 rows affected (0.00 sec)

Step 5: Assign ROLES to Users

```
mysql> GRANT 'admin_role' TO 'admin1'@'localhost';
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> GRANT 'faculty_role' TO 'faculty1'@'localhost';
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> GRANT 'student_role' TO 'student1'@'localhost';
```

Query OK, 0 rows affected (0.00 sec)

Activate Role (Mandatory)

```
mysql> SET DEFAULT ROLE ALL TO
```

```
-> 'admin1'@'localhost',
```

```
-> 'faculty1'@'localhost',
```

```
-> 'student1'@'localhost';
```

Query OK, 0 rows affected (0.01 sec)

Step 6: Check Granted Privileges (Output)

```
mysql> SHOW GRANTS FOR 'faculty1'@'localhost';
```

```
+-----+
| Grants for faculty1@localhost          |
+-----+
| GRANT USAGE ON *.* TO `faculty1`@`localhost`      |
| GRANT `faculty_role`@`%` TO `faculty1`@`localhost` |
```

+-----+

2 rows in set (0.00 sec)

Step 7: Login / Switch Between Users (Windows)

mysql> exit

Bye

C:\Users\Administrator>mysql -u student1 -p

Enter password: *****

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 13

Server version: 8.4.8 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

Step 8: Test Access (Input & Output)

Student User

mysql> SELECT * FROM college_db.student;

```

+-----+-----+-----+
| roll_no | name | marks |
+-----+-----+-----+
|    1 | Amit |   85 |
|    2 | Neha |   90 |
|    3 | Ravi |   78 |
+-----+-----+-----+

3 rows in set (0.00 sec)

```

```
mysql> INSERT INTO college_db.student VALUES (4,'Pooja',88);

ERROR 1142 (42000): INSERT command denied to user 'student1'@'localhost' for table 'student'
```

Step 9: REVOKE Privileges

```
mysql> REVOKE UPDATE ON college_db.student FROM 'faculty_role';

ERROR 1142 (42000): UPDATE, GRANT command denied to user 'student1'@'localhost' for table 'student'

mysql> SHOW GRANTS FOR 'faculty1'@'localhost';

ERROR 1142 (42000): SELECT command denied to user 'student1'@'localhost' for table 'user'

mysql> DROP ROLE 'student_role';

ERROR 1227 (42000): Access denied; you need (at least one of) the CREATE USER, DROP ROLE privilege(s) for this operation
```

Step 10: Drop Role / User (Optional)

```
mysql> DROP USER 'student1'@'localhost';
```

c) Transaction Control (TCL): Apply START TRANSACTION, COMMIT, ROLLBACK, and SAVEPOINT commands to manage transactions.

```
mysql> create database Bank;
```

Query OK, 1 row affected (0.01 sec)

```
mysql> use Bank;
```

Database changed

```
mysql> CREATE TABLE Account (Acc_No INT PRIMARY KEY,Acc_Holder  
VARCHAR(50),Balance INT);
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> INSERT INTO Account VALUES(101, 'Amit Patil', 10000),(102, 'Neha Sharma',  
15000);
```

Query OK, 2 rows affected (0.01 sec)

Records: 2 Duplicates: 0 Warnings: 0

```
mysql> select * from Account;
```

```
+-----+-----+-----+  
| Acc_No | Acc_Holder | Balance |  
+-----+-----+-----+  
| 101 | Amit Patil | 10000 |  
| 102 | Neha Sharma | 15000 |  
+-----+-----+-----+
```

2 rows in set (0.00 sec)

```
mysql> START TRANSACTION;
```

Query OK, 0 rows affected (0.00 sec)

```
mysql> UPDATE Account SET Balance = Balance - 2000 WHERE Acc_No = 101;
```

```
Query OK, 1 row affected (0.00 sec)
```

```
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> UPDATE Account SET Balance = Balance + 2000 WHERE Acc_No = 102;
```

```
Query OK, 1 row affected (0.00 sec)
```

```
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SAVEPOINT transfer_done;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> select * from Account;
```

```
+-----+-----+-----+
| Acc_No | Acc_Holder | Balance |
+-----+-----+-----+
| 101 | Amit Patil | 8000 |
| 102 | Neha Sharma | 17000 |
+-----+-----+-----+
```

```
2 rows in set (0.00 sec)
```

```
mysql> UPDATE Account SET Balance = Balance - 5000 WHERE Acc_No = 101;
```

```
Query OK, 1 row affected (0.00 sec)
```

```
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> ROLLBACK TO transfer_done;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> select * from Account;
```

```
+-----+-----+-----+
| Acc_No | Acc_Holder | Balance |
+-----+-----+-----+
|  101 | Amit Patil |  8000 |
|  102 | Neha Sharma | 17000 |
+-----+-----+-----+

2 rows in set (0.00 sec)
```

mysql> COMMIT;

Query OK, 0 rows affected (0.01 sec)

mysql> select * from Account;

```
+-----+-----+-----+
| Acc_No | Acc_Holder | Balance |
+-----+-----+-----+
|  101 | Amit Patil |  8000 |
|  102 | Neha Sharma | 17000 |
+-----+-----+-----+

2 rows in set (0.00 sec)
```